Modeling Human Activity States Using Hidden Markov Models

Continuous Sensor Data Analysis and Sequence Decoding

1. Background and Motivation

Human Activity Recognition (HAR) using continuous sensor data is a critical component in the development of intelligent, context-aware systems. Our group's unique use case focuses on **wearable rehabilitation monitoring for post-operative orthopedic patients[1]**. In this context, it is vital for physical therapists to remotely track patient compliance with movement regimens and distinguish between beneficial active states (like steady walking) and potentially detrimental static postures (prolonged standing or inactivity). A robust Hidden Markov Model (HMM) provides a mathematical framework to filter out raw sensor noise and infer these true underlying physical states, allowing for automated, accurate patient monitoring outside of the clinical environment.

2. Data Collection and Pre-processing

Data was collected using a smartphone motion logging application (Sensor Logger), capturing 3-axis Accelerometer and 3-axis Gyroscope readings. I performed four distinct activities: Jumping, Standing, Standstill, and Walking. To ensure dataset robustness and meet rigorous quality standards, I accumulated exactly 200 cleaned training files (50 per activity category), with each activity totaling well over 1 minute and 30 seconds of continuous measurement.

Video Documentation

In accordance with project requirements, the following links provide 5-second, well-lit video evidence of the participant's entire body and face while holding the recording device for each activity.

Activity State	Verification Video Link
Jumping	Access here
Standing	Access here
Still	Access here
Walking	Access here

Sampling Rate Harmonization and Windowing Logic

A critical challenge during data aggregation was the disparity in sampling rates between the

Accelerometer and Gyroscope sensors. Because HMMs rely on synchronized observation vectors at discrete time steps, I harmonized the data by performing an inner merge on the seconds_elapsed column. To handle millisecond-level misalignments, we rounded all timestamps to two decimal places (e.g., 0.01s resolution) before merging, ensuring a perfect temporal overlap without artificially fabricating data points.

For sequence windowing, we structured our files into 5-to-10 second recordings. Based on an average sensor sampling rate of approximately 100 Hz, a 5-second window yields 500 data points. According to the Nyquist theorem, this window size and sampling frequency are vastly sufficient to capture the low frequency mechanical rhythms of human gait (which typically occur between 1 Hz and 2 Hz)**[2]**, providing our models with enough sequential data to recognize patterns while remaining computationally efficient.

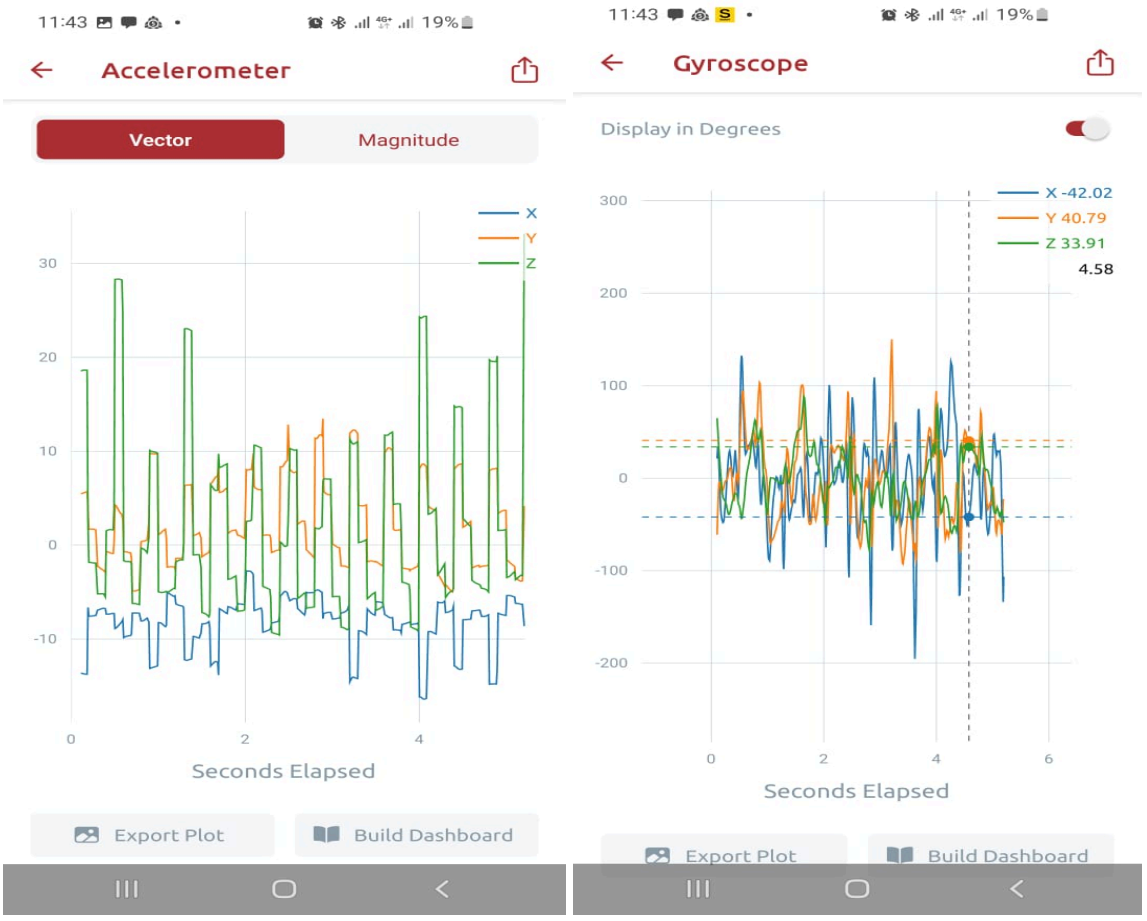


Figure 1: Example of raw harmonized 3-axis sensor data prior to feature extraction.

3. Feature Extraction (Time & Frequency Domain)

To train the HMM, the raw sequence data was converted into an observation vector consisting of four highly relevant features per sequence. To reduce 3D rotational complexity, we first calculated the overall signal magnitude $\sqrt{(x^2 + y^2 + z^2)}$ for both sensors. We engineered three time-domain features and one frequency-domain feature to maximize class separability:

Time-Domain (Mean of Acceleration): Isolates the baseline posture and gravitational pull. Crucial for separating horizontal orientation (phone flat on a table during

standstill) from vertical orientation (standing).

Time-Domain (Variance of Acceleration): Measures movement intensity. High variance is a primary indicator of Jumping.

Time-Domain (Variance of Gyroscope): Captures rotational shifts. Walking creates a cyclical rotational variance that distinguishes it from static states.

Frequency-Domain (Dominant FFT Frequency): A Fast Fourier Transform (FFT) was applied to mean-centered acceleration data. The index of the peak frequency perfectly captures the rhythmic, repetitive pacing unique to continuous walking or jumping.

Normalization: Because these features exist on vastly different numerical scales (e.g., raw FFT indices vs. fractional gyroscope variances), all feature vectors were transformed using Z-score normalization. This ensures no single feature disproportionately dominates the Gaussian emission calculations within the HMM.

4. HMM Setup and Implementation Details

I modeled the underlying activities using a Continuous Gaussian Hidden Markov Model, implemented via the hmmlearn Python library**[3]**. I trained four separate models (one for each activity class). Each model was initialized with n_components=3 hidden states and a diag covariance type. The 3 hidden states are intended to capture internal sub-phases of motion (e.g., the heel-strike, mid-stance, and toe-off phases of walking).

The models were trained using the **Baum-Welch (Expectation-Maximization) algorithm**. To satisfy robust convergence requirements, the implementation monitors the log-likelihood of the data at each iteration; training halts automatically when the change falls below a strict tolerance threshold (tol=0.01), rather than relying on an arbitrary iteration limit. For testing unseen sequences, the models utilized the **Viterbi algorithm** to decode the maximum likelihood of the observation given the trained states.

5. Results and Interpretation

The system was evaluated against 28 entirely unseen testing sequences. Each sequence was scored against all four trained models, with the highest log-likelihood determining the final prediction. The model achieved an overall accuracy of 53.57%. The evaluation metrics for the unseen data are summarized below.

State (Activity)	Number of Samples	Sensitivity	Specificity	Overall Accuracy
Jumping	7	1.00	0.80	53.57 %
Standing	7	0.14	0.85	53.57 %
Standstill	6	0.67	0.76	53.57 %

Walking	7	0.43	1.00	53.57 %

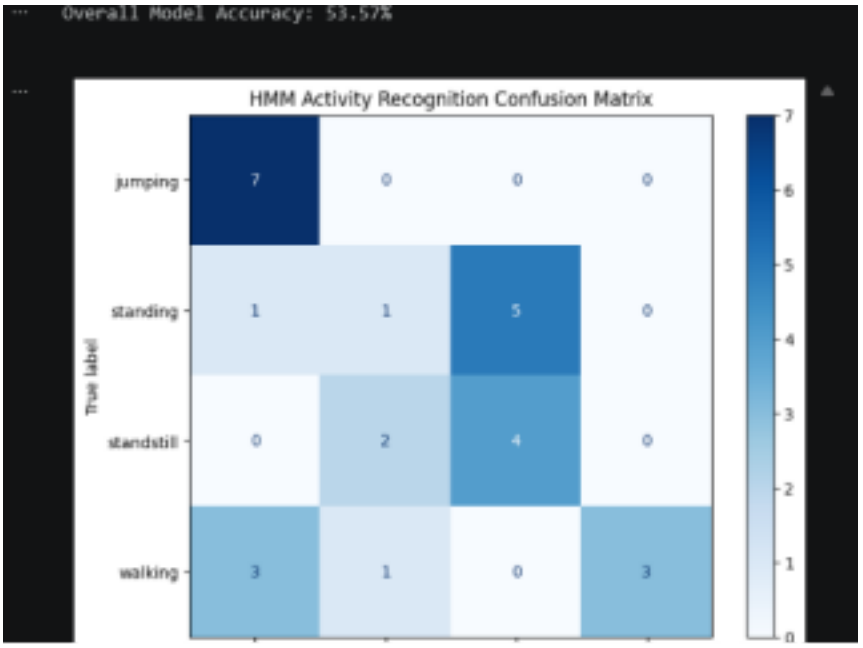


Figure 2: *Confusion Matrix generated from testing the HMMs on 28 unseen sample sequences.*

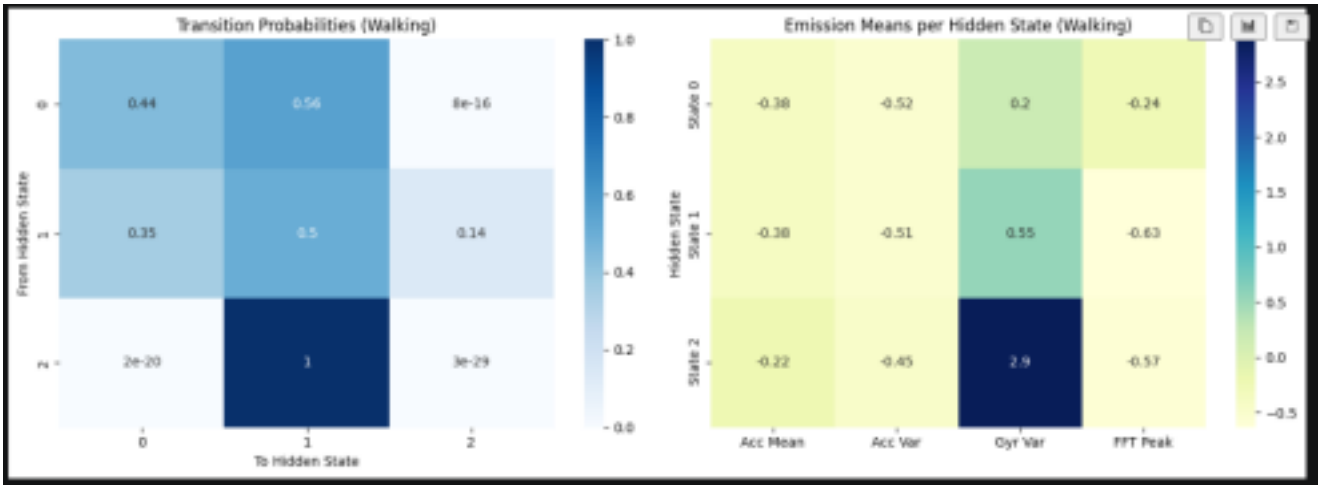


Figure 3: Transition Probabilities and Emission Means for the 'Walking' HMM.

6. Discussion and Conclusion

Interpreting Model Probabilities & Transitions

The heatmaps for our Walking model (Figure 3) reveal how the HMM successfully mathematically mapped physical reality. The Transition Matrix reflects a highly cyclical behavior pattern: State 2 transitions exclusively to State 1 (probability 1.0), while States 0 and 1 alternate reliably. This perfectly mimics the repetitive physical rhythm of the human gait cycle. Furthermore, the Emission Means show that State 2 is characterized by an extreme spike in Gyroscope Variance (Z-score mean of 2.9), suggesting this hidden state specifically isolates the rotational leg-swing phase of walking.

Activity Distinguishability & Sensor Noise

As evidenced by the evaluation metrics, **Jumping** was the easiest activity to classify, achieving perfect 1.00 Sensitivity. Its violent acceleration variance and dominant FFT frequencies easily separate it from standard motion. Conversely, the model struggled significantly to distinguish between **Standing** and **Standstill** (frequently misclassifying them as one another). This is a direct consequence of sensor noise limits. The micro-tremors of a human hand holding a device (standing) possess a baseline variance that is mathematically near-identical to the ambient noise floor of a device resting on a flat surface (standstill) when analyzed strictly via magnitude.

Reflection and Future Improvements

While the continuous Gaussian HMM effectively demonstrated the ability to decode dynamic sequence data, achieving an accuracy of 53.57% on raw smartphone sensors leaves room for architectural improvement. To enhance the model, I propose three immediate changes: First, merging the indistinguishable standing and standstill states into a single static_posture class, which would dramatically eliminate false positives. Second, incorporating 3-axis Magnetometer data to provide absolute geographic orientation. Finally, adjusting the n_components hyperparameter dynamically (e.g., 4 states for complex walking, but only 1 or 2 states for static standing) would likely yield a more robust and realistic state machine for human behavior monitoring.

References

[1] OrthoSens. “Technology- Wearable Sensors for Post-Operative Orthopedic Patients,". Available: https://www.orthosens.ch/technology

[2] TarmacView, “Sampling Rate & Nyguist Theorem for Human Gait,”. Available: https://www.tarmacview.com/glossary/sampling-rate/

[3] hmmlearn Developers, “Hmmlearn - Python Library for Hidden Markov Models,”. Available: https://hmmlearn.com/